

FIT5032 Assessed Lab 5 Report

Library Vue Project

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Implementation Summary

This report documents the completed Lab 5 work for the Library Vue project. The implementation adds password confirmation validation, Vue DevTools evidence, Vue Router navigation, route protection, and a simple login/logout flow.

Requirement	Status	Evidence
Password confirmation	Pass	Confirm password field, mismatch validation, submit-time validation.
Vue DevTools	Pass	HomeView selected and formData expanded in Figure 2.
Vue Router	Pass	Home, About, Login, and Access Denied routes configured.
Route guard	Pass	Protected About route redirects unauthenticated users to Login.
Login/logout	Pass	Demo credentials student / fit5032; logout clears auth state.
Build	Pass	npm run build completed successfully; only bundle-size warning.

Task 5.1: Password Confirmation Validation

The form now includes a Confirm password field bound to `formData.confirmPassword`. Validation displays an error when the confirmation is empty or does not match the password. The check also runs during submit so invalid data cannot bypass blur validation.

W5. Library Regi Form

Let's build some more advar
into our form.

Username

Yilian

Gender

Female

Password

.....

Confirm password

.....

Passwords do not match.

Australian Resident?

Yes

Figure 1. Password confirmation validation shows a mismatch error.

Task 5.1: Vue DevTools Evidence

Vue DevTools was opened while the application was running. The Components panel shows HomeView selected and formData expanded with the current form values.

W5. Library Registration Form

Let's build some more advanced features into our form.

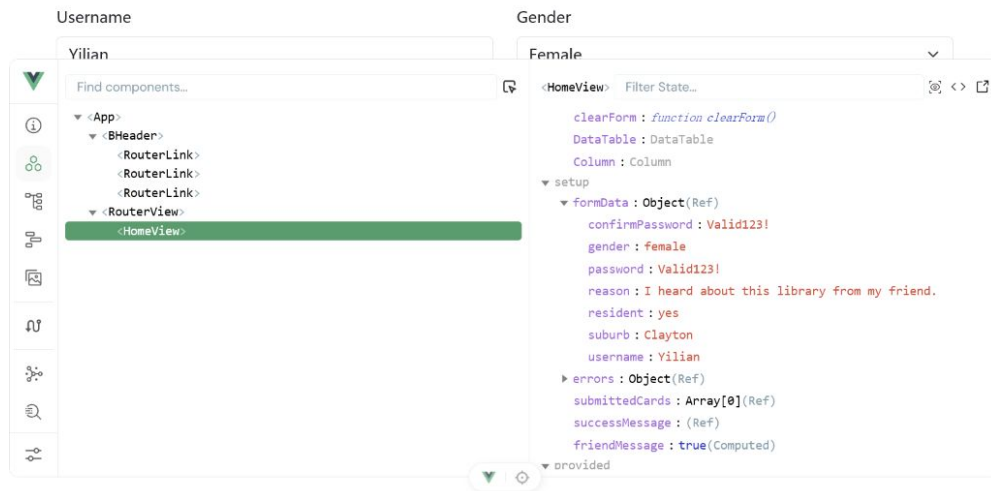


Figure 2. Vue DevTools shows HomeView selected with expanded formData.

Task 5.2: Vue Router and Protected Navigation

Vue Router was configured with Home, About, Login, and Access Denied views. The About route is protected with `meta.requiresAuth` and redirects unauthenticated users to Login with `redirect` and `denied` query parameters.

src/router/index.js

```
import { createRouter, createWebHistory } from 'vue-router'
import { isAuthenticated } from '../auth'
import HomeView from '../views/HomeView.vue'
import AboutView from '../views/AboutView.vue'
import LoginView from '../views/LoginView.vue'
import AccessDeniedView from '../views/AccessDeniedView.vue'

const routes = [
  {
    path: '/',
    name: 'home',
    component: HomeView
  },
  {
    path: '/about',
    name: 'about',
    component: AboutView,
    meta: {
      requiresAuth: true
    }
  },
  {
    path: '/login',
    name: 'login',
    component: LoginView
  },
  {
    path: '/access-denied',
    name: 'access-denied',
    component: AccessDeniedView
  }
]

const router = createRouter({
  history: createWebHistory(import.meta.env.BASE_URL),
  routes
})

router.beforeEach((to) => {
  if (to.meta.requiresAuth && !isAuthenticated.value) {
    return {
      name: 'login',
      query: {
        redirect: to.fullPath,
        denied: '1'
      }
    }
  }

  if (to.name === 'login' && isAuthenticated.value) {
    return { name: 'about' }
  }

  return true
})

export default router
```

src/App.vue

```

<template>
<div class="app-shell">
  <BHeader />

  <main class="container py-5">
    <router-view />
  </main>
</div>
</template>

<script setup>
import BHeader from './components/BHeader.vue'
</script>

<style>
body {
background-color: #f6f8fb;
}

.app-shell {
min-height: 100vh;
}
</style>

```

src/components/BHeader.vue

```

<template>
<header class="navbar navbar-expand-lg navbar-dark bg-primary shadow-sm">
<div class="container">
  <router-link class="navbar-brand fw-semibold" to="/">
  NoMash Library
</router-link>

  <nav>
<ul class="navbar-nav flex-row gap-2 align-items-center">
<li class="nav-item">
  <router-link class="nav-link" active-class="active" to="/">
  Home
</router-link>
</li>
<li class="nav-item">
  <router-link class="nav-link" active-class="active" to="/about">
  About
</router-link>
</li>
<li v-if="!isAuthenticated" class="nav-item">
  <router-link class="nav-link" active-class="active" to="/login">
  Login
</router-link>
</li>
<li v-else class="nav-item">
  <button class="btn btn-outline-light btn-sm" @click="handleLogout">
  Logout
</button>
</li>
</ul>
</nav>
</div>
</header>
</template>

<script setup>
import { useRouter } from 'vue-router'
import { isAuthenticated, logout } from '../auth'

const router = useRouter()

const handleLogout = () => {
  logout()
  router.push('/login')
}
</script>

```

Key HomeView Implementation

The following excerpts show the main reactive state, computed friend message, confirmation password validation, submit handling, and PrimeVue DataTable evidence retained from Lab 4.

HomeView reactive state and computed message

```
import Column from 'primevue/column'

const emptyForm = () => ({
  username: '',
  password: '',
  confirmPassword: '',
  resident: '',
  reason: '',
  gender: '',
  suburb: 'Clayton'
})

const emptyErrors = () => ({
  username: null,
  password: null,
  confirmPassword: null,
  resident: null,
  gender: null,
  reason: null
})

const formData = ref(emptyForm())
const errors = ref(emptyErrors())
const submittedCards = ref([])
const successMessage = ref('')

const friendMessage = computed(() =>
  /\bfriend\b/i.test(formData.value.reason)
)
```

HomeView confirm password validation

```
const validateConfirmPassword = (blur = true) => {
  if (!formData.value.confirmPassword) {
    if (blur) errors.value.confirmPassword = 'Please confirm your password.'
    return false
  }

  if (formData.value.confirmPassword !== formData.value.password) {
    if (blur) errors.value.confirmPassword = 'Passwords do not match.'
    return false
  }

  errors.value.confirmPassword = null
  return true
}

const handlePasswordInput = () => {
  validatePassword(false)

  if (formData.value.confirmPassword || errors.value.confirmPassword) {
    validateConfirmPassword(true)
  }
}

const handleConfirmPasswordInput = () => {
  if (errors.value.confirmPassword) {
    validateConfirmPassword(true)
  }
}
```

HomeView submit logic

```

const validateForm = () => {
  const checks = [
    validateName(true),
    validatePassword(true),
    validateConfirmPassword(true),
    validateResident(true),
    validateGender(true),
    validateReason(true)
  ]

  return checks.every(Boolean)
}

const submitForm = () => {
  successMessage.value = ''

  if (!validateForm()) return

  submittedCards.value.push({
    username: formData.value.username.trim(),
    password: formData.value.password,
    resident: formData.value.resident,
    gender: formData.value.gender,
    suburb: formData.value.suburb.trim() || 'Clayton',
    reason: formData.value.reason.trim()
  })
}

```

HomeView DataTable template

```

</div>
</div>
</div>

<section class="datatable-section mt-5" v-if="submittedCards.length">
  <h2 class="h4 mb-3">Submitted User Information</h2>
  <DataTable
    :value="submittedCards"
    stripedRows
    showGridlines
    responsiveLayout="scroll"
    class="p-datatable-sm"
  >
    <Column field="username" header="Username" sortable />
    <Column field="password" header="Password" />
    <Column header="Australian Resident">
      <template #body="{ data }">
        {{ formatResident(data.resident) }}
      </template>
    </Column>
    <Column header="Gender">
      <template #body="{ data }">
        {{ formatGender(data.gender) }}
      </template>
    </Column>
  </DataTable>

```

Browser Evidence

W5. Library Registration Form

Let's build some more advanced features into our form.

Username

Gender

Please select 

Password

Confirm password

Australian Resident?

Please select 

Suburb

Clayton

Reason for joining



Figure 3. Home page loads successfully on localhost.

About Our Library

Welcome to our digital library! We're dedicated to providing a vast collection of books and resources to our community.

This protected page is available after login for Lab 5 secure routing.



Figure 4. About page content after successful authentication.

Login

Access denied. Please log in to continue.

Demo credentials: **student / fit5032**. This is for Lab demonstration only and must not be used in production.

Username

Password

Login



Figure 5. Unauthenticated About access redirects to Login.

Login

Access denied. Please log in to continue.

Demo credentials: **student / fit5032**. This is for Lab demonstration only and must not be used in production.

Username

Password

Login



Figure 6. Login page shows access denied guidance.

About Our Library

Welcome to our digital library! We're dedicated to providing a vast collection of books and resources to our community.

This protected page is available after login for Lab 5 secure routing.



Figure 7. Valid login reaches the protected About page.

NoMash Library

[Home](#) [About](#) [Login](#)

Login

Demo credentials: **student / fit5032**. This is for Lab demonstration only and must not be used in production.

Username

Password

Login



Figure 8. Logout returns the user to Login and clears authenticated navigation.

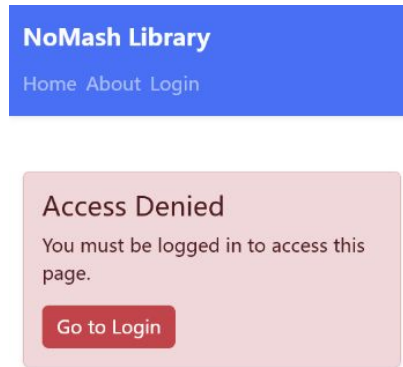


Figure 9. Access Denied route displays the protected-route warning.

Test Results

Test	Result	Notes
Password mismatch	Pass	Displayed Passwords do not match.
Empty confirm password submit	Pass	Displayed Please confirm your password.
Reason includes friend	Pass	Displayed Great to have a friend in green text.
Clear form	Pass	Reset fields, errors, success message, and suburb default.
Unauthenticated About	Pass	Redirected to /login?redirect=/about&denied=1.
Valid login	Pass	student / fit5032 navigated to /about.

Test	Result	Notes
Logout	Pass	Authentication state cleared and About became protected again.
Build	Pass	npm run build completed with no blocking errors.

Conclusion

All Lab 5 requested features have been implemented and verified. The report includes the required Vue DevTools evidence, route protection screenshots, and the key source code used to complete the lab.